

Software Engineer specializing in scalable test infrastructure, developer tooling, and distributed automation systems. Conceived and led the architecture of Deltek’s modular Playwright platform, adopted by 30+ automation engineers across 13 QA teams. Built claims reconciliation, query compilation, and database tooling for Zealie’s healthcare revenue-cycle operations. Formally recognized by client engineering leadership for platform development and adoption.

SKILLS & TECHNOLOGIES

Languages & Core	TypeScript, JavaScript (Node.js), Python, SQL (T-SQL, MySQL 8), Bash, PowerShell
Test Platforms & Automation	Playwright, RESTful API Automation, Schema Validation (Zod), Nightwatch.js, ESLint
CI/CD & Platform Orchestration	Azure DevOps YAML Pipelines, Multi-Shard Parallelization, Linux, Git
Developer Tooling & Security	Model Context Protocol (MCP), AST Codemods (jscodeshift), Query Optimization, mTLS, OS Keyring

PROFESSIONAL EXPERIENCE

IndiaNIC — Senior QA Engineer / SDET

Sep 2022 – Present | Hybrid

Selected client engagements through IndiaNIC: **Deltek** (test infrastructure and developer platform); **Zealie** (backend data systems and platform engineering).

[Client: Deltek Vantagepoint — Test Infrastructure & Developer Platform]

Client context: Project-based ERP and financial management platform for architecture, engineering, and consulting firms.

- **0-to-1 Test Platform Architecture:** Designed Deltek’s 5-layer modular Playwright platform from scratch — building the session engine, dependency injection container, type-safe REST API client layer, and Zod validation layers across 18 ERP modules; authored 340+ framework validation tests and established 450+ baseline smoke suites, and bootstrapped adoption via a DOM crawler generating 422 ERP and 182 SSRS Page Objects, enabling 13 QA teams to contribute 1,490+ feature tests.
- **Distributed CI/CD & Orchestration:** Built a reusable Azure DevOps pipeline architecture powering automated daily scheduled suites and release gates, replacing manual pipeline runs; orchestrated parallel agent execution across a matrix of 12 database configurations, running the 2,300-test suite in ~60m of wall-clock time with merged shard reports and a custom TRX reporter preserving retry failure diagnostics.
- **Metadata-Driven Test Data Engine:** Built a metadata-driven test data engine resolving recursive foreign-key dependency graphs across 32 field types and self-referential cycles via ancestor tracking; enforced worker-scoped key isolation to prevent test-data collisions during dependency resolution across 26 parallel workers, paired with automated reverse-topological LIFO teardown.
- **Framework Evolution & AST Codemods:** Built idempotent jscodeshift codemods to migrate framework APIs; built ESLint autofix rules, reviewed core-framework PRs as a branch-policy reviewer, and trained automation engineers on framework architecture and coding standards.
- **Developer Platform & AI Workflows:** Built a Playwright agent skill system that codified framework architecture, design decisions, and code-review standards, alongside an in-house MCP server extending AI assistants into live ERP environments — using file-based schema retrieval and jq subtree extraction to avoid exhausting the LLM context window during test synthesis.

Client context: Healthcare revenue cycle management (RCM) platform serving 300+ behavioral health facilities.

- **Claims Reconciliation & Discrepancy Analysis:** Built a reconciliation pipeline across CollaborateMD and internal RCM ledgers using normalized 12-field exact compound keys, Institutional vs. Professional claim-type rules, and bucketed similarity-based candidate matching; generated discrepancy sheets linking directly to source records for operator review, shrinking facility reconciliation and migration timelines from over a month to three days.
- **Dynamic Query Compiler & Hybrid AI Workflows:** Built a recursive query compiler in TypeScript that translates MongoDB-style JSON filter trees (\$and, \$or, \$in, \$like) into parameterized MySQL predicates against an allowlisted column schema; used Gemini in Google Sheets to convert unstructured plain-English audit requests into validated JSON expressions, enabling non-technical compliance auditors to execute dynamic medical-record audits without writing SQL.
- **Productivity Analytics & Product Prototyping:** Built a workforce productivity engine computing 15 operational metrics across 8 targeted SQL queries per specialist, including claim touches, co-toucher splits, and payment attribution; reduced JDBC result-access overhead using server-side JSON aggregation (JSON_ARRAYAGG) on a MySQL replica to retrieve one structured result per query for client-side processing. The prototype was subsequently productized into Zion’s enterprise web application.
- **Prefix Discovery & Transactional Migrations:** Built an automated insurance prefix discovery pipeline that mined prefix-to-payer mappings directly from the Blue Cross Blue Shield clearinghouse website, distributing 46,656 candidate lookups across seven daily buckets using a deterministic modulo-7 hash; built a five-stage migration workflow (Discover, Snapshot, Transform, Prepare, Run) with per-patient SQL transactions, generated-ID propagation, and reverse-order compensating undo statements.
- **Serverless Privilege Gateway & Reporting Infrastructure:** Built an intermediary gateway service enabling non-technical operations staff to run database reports from Google Sheets without direct database credentials; routed requests through an execution boundary that translated unprivileged user calls into authorized service-identity queries, using asynchronous trigger queues to decouple long-running SQL executions and write formatted datasets back to destination sheets.
- **Developer Tooling & AI Workflows:** Built a Python CLI using uv, Click, and Pydantic managing database access across dev, QA, and production environments, with environment-specific TLS client certificates, OS keyring credential storage, and read-only session defaults; added an interactive terminal (TTY) interlock that rejects automated non-interactive execution against production through the CLI and requires explicit confirmation, reducing accidental execution risk during developer and AI-assisted database inspection.

OPEN SOURCE CONTRIBUTIONS

- **Microsoft Playwright:** Investigated and submitted root-cause analysis with reproducible regression test cases for custom reporter AggregateError serialization defects (resolved upstream in v1.61 milestone, [Issue #40856](#)); authored browser runtime workaround for CapsLock keyboard modifier state handling under CDP/BiDi ([Issue #16214](#)).
- **Nightwatch.js:** Isolated and submitted issue reports with reproducible test cases for global hook execution-order lifecycle defects and element locate strategy failures ([Issue #3847](#), [Issue #3530](#)).
- **UnJS (unpdf):** Resolved dual-package type resolution defects in TypeScript dual ESM/CJS declaration exports ([PR #56](#)).

EDUCATION

M.S. in Machine Learning & AI — Liverpool John Moores University	2021 – 2022
Postgraduate Diploma in Machine Learning & AI — IIIT Bangalore	2020 – 2021
Master of Computer Applications — Amity University, Noida	2016 – 2019
Bachelor of Science — Ewing Christian College, Prayagraj	2016